Q Will I know right away if my baby has cystic fibrosis (CF)?

About one in 10 babies with CF have a condition called meconium ileus at, or shortly after, birth, which causes a bowel blockage and requires surgery. Other symptoms of CF usually emerge in the first year of life, so it may not be obvious immediately that your baby has CF. However, most cases of CF are now diagnosed shortly after birth since CF is one of the conditions screened for with the heel prick test (see p.250), done within the first few days of life. Signs of CF include difficulty breathing, a raspy, hacking cough that doesn't go away, and coughing up a lot of phlegm. If the condition affects your baby's pancreas, she'll be unable to produce the right level of enzymes to break down her food sufficiently, and her digestive juices will be too thick to absorb nutrients properly. As a result, she may show signs of malnutrition, such as having pale skin, a thin and small frame, and a distended tummy, and may pass fatty-looking stools. Her skin may taste salty, because your baby's sweat has a high concentration of salt.

Q Can cystic fibrosis (CF) be treated?

There is no cure for CF. However, although it is a life-limiting condition, advances in medicine over recent decades mean that life expectancy has dramatically improved, and now more than 80 percent of babies born with CF will live into their 50s. Treatment involves relieving symptoms, and is tailored to the individual's needs. Physical therapy, oxygen therapy, and inhalers can be used to ease breathing, while vitamin and mineral supplements, replacement digestive enzymes, and a high-calorie diet can help to improve nutrition and growth. Your baby will be more vulnerable to infections, so may also be prescribed prophylactic (preventative) antibiotics. In more serious cases, a heart and lung transplant may be needed.

There are certain characteristic effects of CF on the sufferer's body, including, most significantly, impairment of the function of the lungs and pancreas.

» **Damage to the pancreas** can not only cause problems with digestion, but also lead to diabetes since the pancreas is responsible for making insulin, which controls sugar levels.

» **The intestine** can't absorb nutrients properly from food.

» **Increased likelihood of infertility**, particularly for men.

» **Poor nutrition** can lead to osteoporosis later on, especially in women.

» **Sinus infections** are common, as well as polyps (small growths) inside the nose.

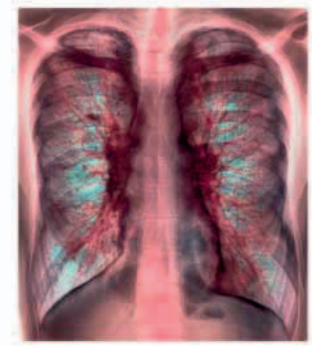
» **The lungs fill with mucus**, which leads to breathing difficulties, repeated chest infections, and increased coughing.

» **Risk of liver damage.**

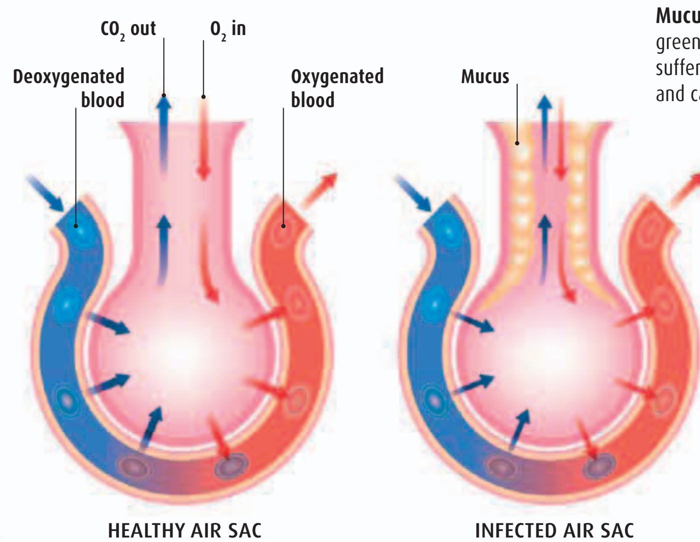
Q What is cystic fibrosis, and how does it affect babies?

Cystic fibrosis (CF) is a condition caused by a faulty gene, called CFTR. The gene needs to be inherited from both parents for someone to suffer from CF. The faulty gene causes a buildup of mucus that in turn affects breathing and feeding.

The condition primarily affects the lungs and digestive system, causing a buildup of mucus in these organs. The faulty gene affects the way in which salt and water move in and out of cells in the body; an excess of salt enters the cells of certain organs, causing an imbalance of salt and water. Eventually too little water remains in the fluid surrounding the affected organ's cells, resulting in a buildup of thick mucus that impairs the organ's function. The condition affects one in 3,700 babies born in the US every year. About 10 million Americans unknowingly carry the faulty gene.



Mucus buildup Here, mucus (in green) can be seen clearly in a CF sufferer's lung, blocking airways and causing difficulty breathing.



Air sacs This illustration shows how the buildup of mucus seen in the lungs of a cystic fibrosis sufferer clogs up the multitude of air sacs (alveoli). This affects the flow of oxygen in the lungs and causes difficulty with breathing.

Q Will cystic fibrosis affect my baby's growth?

This partly depends upon how severely she has the condition. Mutations in the gene that causes cystic fibrosis mean that it varies in its effects. Mild forms of the disease may affect only the lungs, and her pancreas may remain relatively stable. This will mean that she won't suffer from the nutritional deficiencies associated with more severe forms of the disease and is more likely to grow normally. Even in more severe cases, most sufferers are able to take supplementary pancreatic enzymes that help do the job that the pancreas is failing to do. This can help with the breakdown of food and has been shown to greatly improve growth in those with CF.